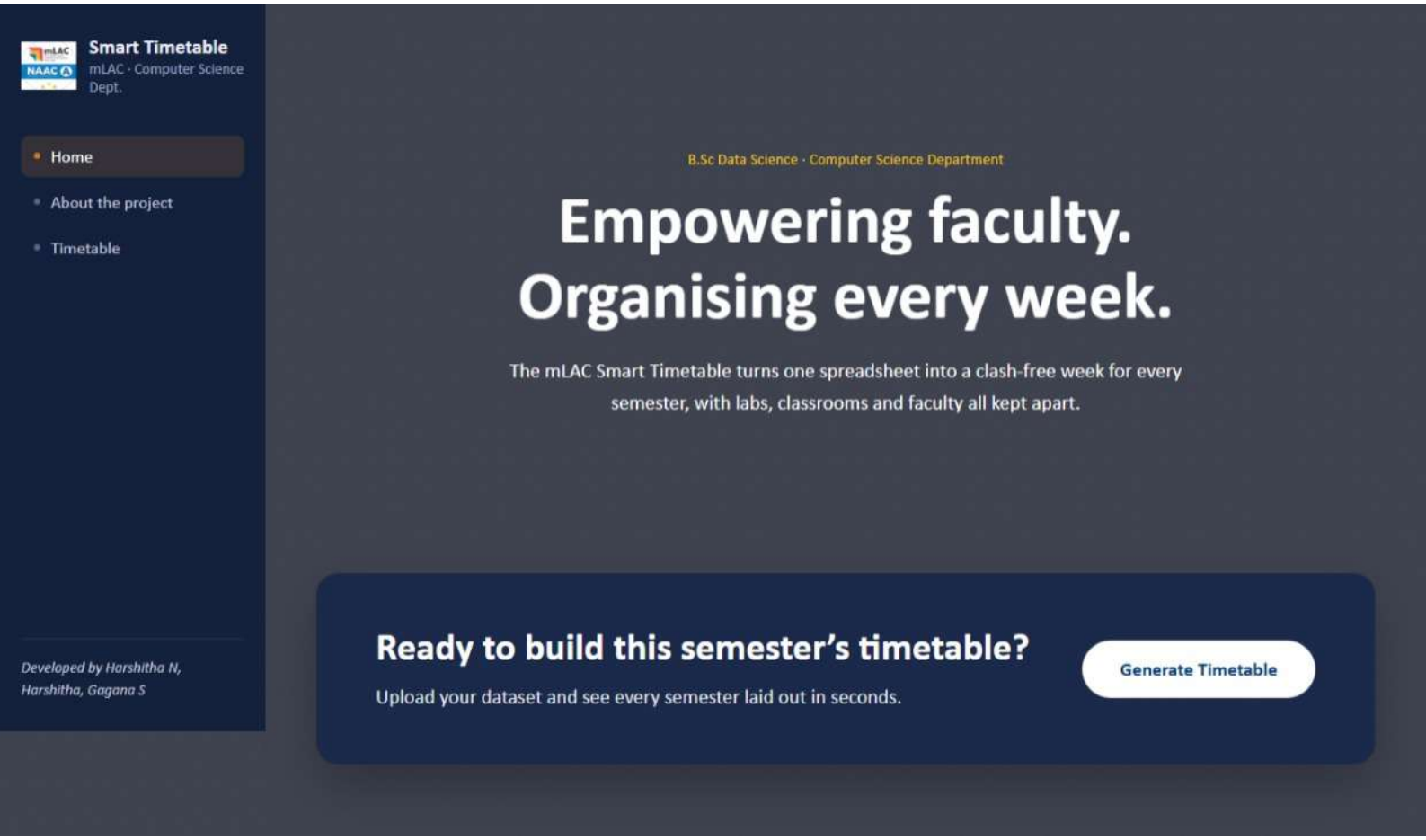

Smart Timetable
mLAC · Computer Science Dept.
Home
About the project
Timetable
Developed by Harshitha N, Harshitha, Gagana S
B.Sc Data Science · Computer Science Department
Empowering faculty. Organising every week.
The mLAC Smart Timetable turns one spreadsheet into a clash-free week for every semester, with labs, classrooms and faculty all kept apart.
Ready to build this semester's timetable?
Upload your dataset and see every semester laid out in seconds.
Generate Timetable

About the project

Smart AI-Based Automatic Timetable Generator

Built for the Department of Computer Science.

What the project does

Preparing a college timetable by hand means juggling faculty, classrooms, laboratories and batches across several semesters. This project does it automatically from a single dataset, and checks that nothing clashes.

How it works

Step	What happens
1. Read the workbook	Academic, Resource, Batches, Faculty and Slot sheets are read and cleaned, and problem rows are flagged.
2. Plan labs, then theory	A Genetic Algorithm places lab blocks first, pairs faculty, then places every theory period.
3. Validate & export	Every rule is re-checked from scratch, then the master and semester timetables are shown and exported to Excel.

Technologies used

Python · Genetic Algorithm · pandas · openpyxl · HTML · CSS · JavaScript

Developed by Harshitha N, Harshitha, Gagana S

Developed by Harshitha N, Harshitha, Gagana T

Step 2

Generated timetable

Theory in amber, labs in teal. All 12 hard-constraint checks passed · no faculty, room or laboratory clashes.

Master timetable

all semesters

DAYS	SEMESTERS	8:50–9:20	9:20–10:10	10:10–11:00	11:00–11:50	11:50–12:40	12:40–1:30	1:30–2:20
MONDAY	1st Sem	Maths SB · Kishore	C Programming SB · Anusha	—	—	—	—	C Programming Lab Batch A · Lab 1 · Anusha & Bindhu 2:00–5:00 Computer Organization Lab Batch B · Lab 2 · Meghana D & Divya 2:00–5:00
	3rd Sem	—	—	R Programming SB · Shilpita	Cyber Security SB · Gayathri Bai	Principles of Data Science SB · Gayathri Bai	—	—
	5th Sem	Data Visualization S10 · Divya	Machine Learning S10 · V. Manjula	Cloud Computing S10 · Sparsha	Project Lab Batch A · Lab 1 · V. Manjula & Bindhu 11:00–5:00 Project Lab Batch B · Lab 2 · Divya & Shilpita 11:00–5:00			
TUESDAY	1st Sem	—	—	Computer Organization SB · Meghana D	Maths SB · Kishore	—	—	—
	3rd Sem	Probability SB · Meghana D & Anusha	Statistics SB · Meghana D & Anusha	—	Principles of Data Science Lab Batch A · Lab 1 · Gayathri Bai & Sparsha 11:00–2:00 R Programming Lab Batch B · Lab 2 · Shilpita & V. Manjula 11:00–5:00			
	5th Sem	Data Visualization S10 · Divya	Big Data Analytics S10 · Shreya	—	—	—	—	Data Visualization Lab Batch A · Lab 1 · Divya & Bindhu 2:00–5:00 Machine Learning Lab Batch B · Lab 2 · V. Manjula & Gayathri Bai 5:00–8:00
WEDNESDAY	1st Sem	Computer Organization Lab Batch A · Lab 1 · Meghana D & Divya 8:00–11:00 C Programming Lab Batch B · Lab 2 · Anusha & Bindhu 8:00–11:00			—	C Programming SB · Anusha	—	—
	3rd Sem	R Programming SB · Shilpita	Principles of Data Science SB · Gayathri Bai	Cyber Security SB · Gayathri Bai	—	—	—	—
	5th Sem	—	Cloud Computing S10 · Sparsha	Business Analysis S10 · Jayasree	Machine Learning S10 · V. Manjula	Data Visualization S10 · Divya	—	—
THURSDAY	1st Sem	Computer Organization S10 · Meghana D	C Programming SB · Anusha	Maths SB · Kishore	—	—	—	—
	3rd Sem	R Programming SB · Shilpita	—	—	—	—	—	Probability & Statistics Lab Batch A · Lab 1 · Meghana D & Anusha 2:00–5:00 Principles of Data Science Lab Batch B · Lab 2 · Gayathri Bai & Sparsha 2:00–5:00
	5th Sem	Machine Learning Lab Batch A · Lab 1 · V. Manjula & Gayathri Bai 8:00–11:00 Data Visualization Lab Batch B · Lab 2 · Divya & Bindhu 8:00–11:00			Big Data Analytics S10 · Shreya	Cloud Computing S10 · Sparsha	—	—
FRIDAY	1st Sem	Office Automation Lab Batch A · Lab 1 · Sparsha & Shilpita 8:00–11:00 Office Automation Lab Batch B · Lab 2 · Bindhu & V. Manjula 8:00–11:00			C Programming SB · Anusha	Computer Organization SB · Meghana D	—	—
	3rd Sem	Principles of Data Science SB · Gayathri Bai	Statistics SB · Meghana D & Anusha	Probability SB · Meghana D & Anusha	—	—	—	—
	5th Sem	Data Visualization S10 · Divya	Big Data Analytics S10 · Shreya	Business Analysis S10 · Jayasree	Machine Learning S10 · V. Manjula	—	—	—
SATURDAY	1st Sem	—	—	Computer Organization SB · Meghana D	Maths SB · Kishore	—	—	—
	3rd Sem	Principles of Data Science SB · Gayathri Bai	R Programming SB · Shilpita	—	R Programming Lab Batch A · Lab 2 · Shilpita & V. Manjula 11:00–5:00 Probability & Statistics Lab Batch B · Lab 2 · Meghana D & Anusha 11:00–2:00			
	5th Sem	Cloud Computing S10 · Sparsha	Big Data Analytics S10 · Shreya	Machine Learning S10 · V. Manjula	—	—	—	Project Lab Batch A · Lab 1 · V. Manjula & Shilpita 2:00–5:00 Project Lab Batch B · Lab 2 · Divya & Anusha 2:00–5:00

Semester-wise view

Semester 1 | Semester 3 | Semester 5

Day	8:00–11:00	8:20–9:20	9:20–10:10	10:10–11:00	11:00–2:00	11:00–11:50	11:50–12:40	12:40–1:30	1:30–2:20	2:00–5:00
Monday	—	Maths SB · Kishore	C Programming SB · Anusha	—	—	—	—	—	—	C Programming Lab Batch A · Lab 1 · Anusha & Bindhu 2:00–5:00 Computer Organization Lab Batch B · Lab 2 · Meghana D & Divya 2:00–5:00
Tuesday	—	—	—	Computer Organization SB · Meghana D	—	Maths SB · Kishore	—	—	—	—
Wednesday	Computer Organization Lab Batch A · Lab 1 · Meghana D & Divya 8:00–11:00 C Programming Lab Batch B · Lab 2 · Anusha & Bindhu 8:00–11:00	—	—	—	—	—	C Programming SB · Anusha	—	—	—
Thursday	—	Computer Organization S10 · Meghana D	C Programming SB · Anusha	Maths SB · Kishore	—	—	—	—	—	—
Friday	Office Automation Lab Batch A · Lab 1 · Sparsha & Shilpita 8:00–11:00 Office Automation Lab Batch B · Lab 2 · Bindhu & V. Manjula 8:00–11:00	—	—	—	—	C Programming SB · Anusha	Computer Organization SB · Meghana D	—	—	—
Saturday	—	—	—	Computer Organization SB · Meghana D	—	Maths SB · Kishore	—	—	—	—

Generated from the uploaded dataset by a Genetic Algorithm (lab placement + theory placement; 335 generations, fitness 0.0269, 20.5s).

Download master & semester timetables (.xlsx)